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K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai

SMART WATER QUALITY MONITORING
MINI PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements of the Degree of
Bachelor of Technology in Electronics and Telecommunication Engineering

By

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Mohd. Gibran Ulde (A-55)
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CERTIFICATE

*This is to certify that the project entitled " SMART WATER QUALITY MONITORING" is a bonafide work of students Ansh Taralekar, Gibran Ulde, Harsh Saraiya, Vraj Shah. submitted to the K J Somaiya Institute of Technology (An Autonomous Institute) in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Electronics and Telecommunication Engineering***.**

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MINI PROJECT APPROVAL

for

Bachelor of Technology in Electronics and

Telecommunication Engineering

This Project report entitled

"SMART WATER QUALITY MONITORING"

By

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Supervisors:

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Ansh Taralekar

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Vraj Shah

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INTRODUCTION

In the era of IoT and automation, this project uses Raspberry Pi as a low-cost, powerful core unit to build a real-time water quality monitoring system. The goal is to enable scalable, intelligent, and automated water analysis for early pollution detection.

Key Features

1. Raspberry Pi: Central processor for data handling, AI inference, and cloud sync

2. Sensors Used: pH, turbidity, dissolved oxygen, heavy metals (e.g., lead, mercury)

3. Real-Time Monitoring: Continuous tracking of water parameters using IoT sensors

4. Scalable Design: Suitable for rural wells, urban pipelines, and industrial use

Impact

1. Improves public health through early contamination alerts

2. Supports data-driven decisions for sustainable water management

3. Enables data-driven policymaking and smarter resource allocation

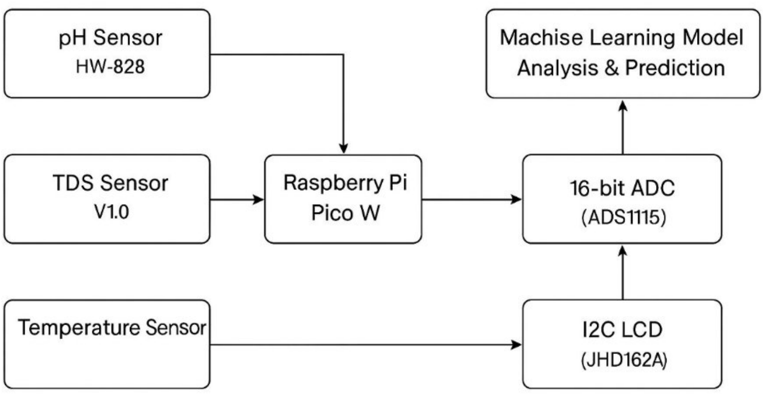

Figure 1.1: Block diagram of Smart Water Quality Monitoring

LITERATURE SURVEY

The current project stage involves real-time monitoring of TDS and pH levels using low-cost sensors integrated with microcontrollers. Literature supports the feasibility and impact of such systems in ensuring water safety.

Sr no.	Article	Author	Gaps
1	Sensor-Based Monitoring	DFRobot Sensor Guides	Limited calibration guidelines for different water types
2	IoT Integration	Patel et al. (2021), IJRASET	Lacks cloud scalability and long-term data storage
3	Use in Indian Case Studies	Deshmukhetal. (2020), IJSRD	Insufficient rural network coverage
4	Smart Water Monitoring System	Raj and Babu (2019), IJERT	No AI-based analysis or alerts
5	Wireless Sensor Networks in Water Quality	Hariharan et al. (2017), Procedia CS	High power consumption in continuous sensing
6	Smart Water Quality Using Raspberry Pi	Kavitha et al. (2018), IJPAM	Minimal security in data transmission
7	IoT for Drinking Water Monitoring	Chaitanya et al. (2019), IJRTE	Delay in real-time alerts under poor connectivity
8	Low-Cost Sensor Comparison	Kumar & Sharma (2020), Journal of Water and Health	Limited benchmarking across water sources
9	Rural Area Monitoring Systems	Sharma et al. (2021), IJIRSET	Maintenance challenges in rural deployments
10	Cloud-Based Water Monito	Verma & Gupta (2020), IJERT	Lacks edge computing or offline fallback

11	Urban Water Quality Monitoring	Singh & Mehta (2018), IJRASET	Integration with municipal systems not addressed
12	AI-Assisted Water Monitoring	Patel & Rathi (2020), IJSRD	Needs more diverse datasets for ML training
13	GSM-Enabled Water Monitoring	Nair & Kumar (2017), IJAREEIE	GSM delays and high energy usage
14	Water Quality Monitoring in Agriculture	Desai et al. (2019), IJLTEMAS	Does not account for seasonal water variation
15	Real-Time pH and TDS Monitoring App	Sharma et al. (2018), IJRASET	App not synchronized with cloud dashboards

OBJECTIVE AND PROBLEM STATEMENT

Problem Statement:

Access to clean and safe drinking water remains a critical challenge, with millions affected globally by waterborne diseases caused by undetected contamination. In countries like India, over 70 million people are exposed to hazardous pollutants such as arsenic, fluoride, and industrial waste. Existing water quality assessment methods are often manual, slow, and cost-prohibitive—failing to provide timely detection or widespread coverage. This gap in real-time, efficient water monitoring demands an innovative and scalable solution to ensure safe water access and protect public health.

Objective:

The objective of this project is to design and implement a smart water quality monitoring system that leverages IoT-based sensors for real-time detection of critical water parameters such as pH, turbidity, and heavy metals. By integrating machine learning algorithms, the system aims to provide instant contamination alerts and predictive risk assessments, enabling proactive intervention. This automated and cost-effective solution is intended to reduce reliance on traditional, manual testing methods, thereby improving response times, safeguarding public health, protecting natural water resources, and ensuring widespread access to clean and safe drinking water.

CIRCUIT DIAGRAM

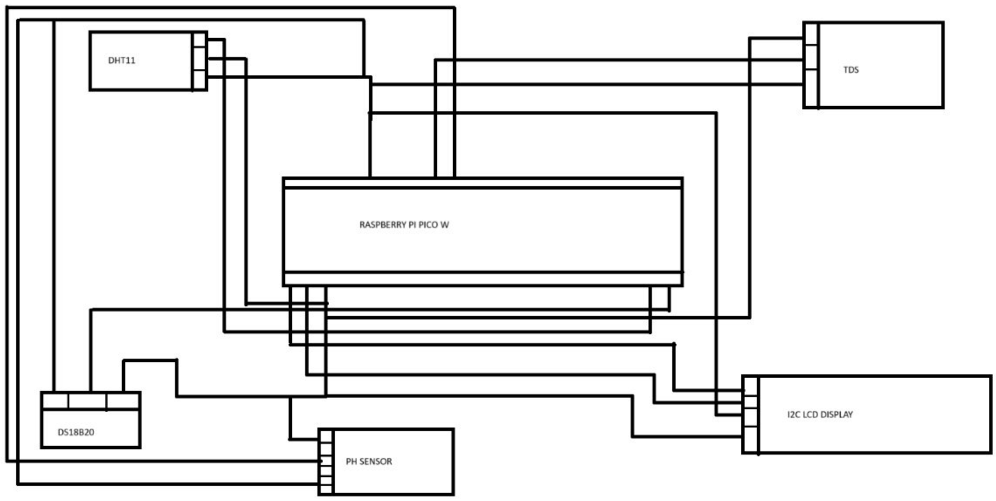

Figure 4.1: Circuit Diagram Of Smart Water Quality Monitoring System

Components:

1. Raspberry Pi Pico W – Microcontroller for data processing and control

2. pH Sensor (HW-828) – Measures acidity/alkalinity of water

3. TDS Sensor (V1.0) – Measures total dissolved solids in water

4. Temperature Sensor – Monitors water temperature (e.g., DS18B20)

5. JHD162A I2C LCD Display – Displays real-time readings and alerts

6. Breadboard & Jumper Wires – For circuit connections

7. Power Supply/USB Cable – To power the Pico W

Connections:

1.DHT11 (Air Temp) - GPIO 15

2.DS18B20 (Water Temp) - GPIO 2

3.TDS Sensor - GPIO 27 (ADC1)

4.I2C LCD (SCL) - GPIO 1 (I2C0 SCL)

5.I2C LCD (SDA) - GPIO 0 (I2C0 SDA)

6.All Sensors - 3.3V or 5V and GND

SENSORS

1.Raspberry Pi Pico W:

Microcontroller: RP2040 (Dual-core ARM Cortex-M0+)

Clock Speed: 133 MHz

RAM: 264 KB SRAM

Flash Memory: 2 MB onboard QSPI flash

Wireless: 2.4 GHz 802.11n Wi-Fi (via Infineon CYW43439 chip)

GPIO Pins: 26 multi-function GPIOs

Power Input: 1.8–5.5V (via USB or VSYS)

Programming Interface: USB (Micro-B) using C/C++ or MicroPython

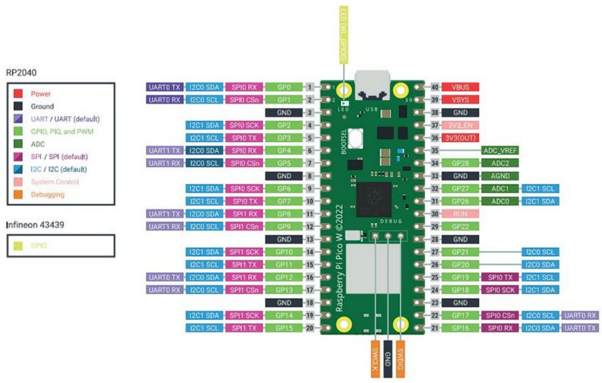

Figure 4.2.a Raspberry Pi Pico W

2. pH Sensor:

Input Voltage: 5V DC

Output Voltage: 0 – 3V analog signal

Measurement Range: pH 0 – 14

Accuracy: ±0.1 pH (at 25°C)

Working Temperature: 0 – 60°C

Connector Type: BNC for probe, 3-pin header for module

Calibration: Manual (using provided potentiometer)

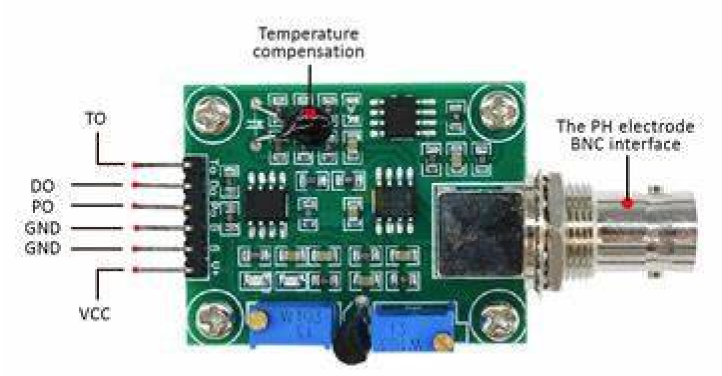

Figure 4.2.b pH sensor module

3. TDS Sensor:

Input Voltage: 3.3 – 5.5V DC

Output Signal: Analog voltage

Measurement Range: 0 – 1000 ppm (parts per million)

Accuracy: ±10% F.S.

Operating Temperature: 0 – 80°C

Probe Type: Waterproof, anti-corrosion

Interface: 3-pin header (VCC, GND, AOUT)

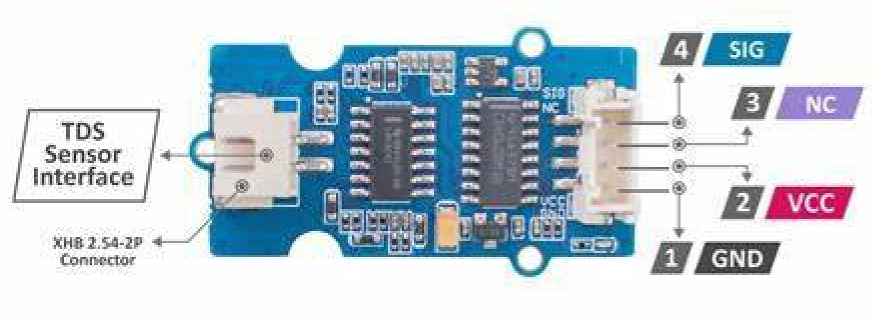

Figure 4.2.c TDS sensor module

4. DHT11 Temperature Sensor:

Sensor Type: Digital Temperature and Humidity

Temperature Range: 0 – 50°C ±2°C

Humidity Range: 20 – 90% RH ±5%

Interface: Single-wire digital

Sampling Rate: 1 Hz (once every second)

Power Supply: 3.3V to 5.5V

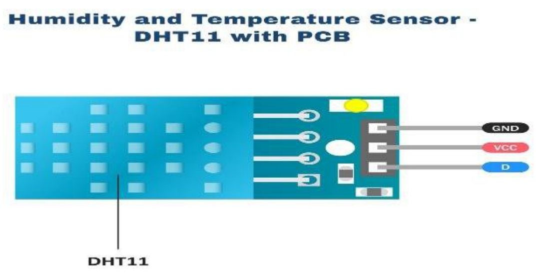

Figure 4.2.d DHT11 External Temperature sensor

5. DS18B20 Temperature Sensor:

Interface: 1-Wire Digital

Temperature Range: -55°C to +125°C

Accuracy: ±0.5°C (from -10°C to +85°C)

Resolution: 9 to 12 bits (configurable)

Power Supply: 3.0V to 5.5V

Waterproof Probe: Sealed stainless steel

Figure 4.2.e DS18B20 Internal Temperature Sensor

6. I2C LCD Display

Display Size: 16 characters × 2 lines

Controller: HD44780 compatible

Interface: I2C (via PCF8574 or similar adapter)

Operating Voltage: 5V DC

Backlight: LED

I2C Address: Typically 0x27 or 0x3F (check with scanner)

Pins: VCC, GND, SDA, SCL

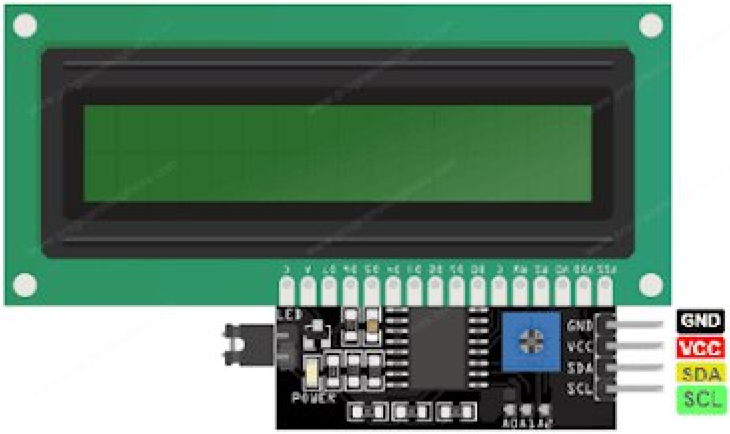

Figure 4.2.f I2C LCD Display

WORKING

The system begins with three main sensors—pH (HW-828), TDS (V1.0), and Temperature—that continuously monitor water quality parameters.

These sensors output analog signals, which are routed to the ADS1115 (16-bit ADC) for high-resolution analog-to-digital conversion, ensuring accurate and noise-resistant readings.

The Raspberry Pi Pico W acts as the central controller, receiving digital data from the ADC and handling all processing, control, and communication tasks.

The Pico W also interfaces with a pre-trained AI model, either onboard or remotely, which performs real-time analysis and predictive evaluation of the water quality.

Based on data analysis, the system can identify abnormal patterns or contamination risks before they become critical.

The processed data is shown on a JHD162A I2C LCD, providing clear, real-time feedback to the user on current water conditions.

This design ensures a low-power, cost-effective, scalable, and intelligent monitoring system suitable for domestic, agricultural, and industrial applications

SIGNIFICANCE OF 2 TEMPERATURE SENSORS:

Enhanced Environmental Intelligence: Simultaneous tracking of air and water temperature enables a deeper understanding of ecological dynamics. For example, elevated water temperatures reduce oxygen solubility, directly impacting aquatic biodiversity.

1. Sensor Data Precision and Calibration: Water temperature significantly influences the accuracy of sensor outputs (e.g., pH and TDS). Incorporating ambient air temperature allows for more reliable compensation algorithms, thereby improving data fidelity.

2. Advanced Predictive Modeling: Dual-temperature monitoring facilitates the development of models that can anticipate seasonal shifts, pollution patterns, or biologically sensitive events like algal blooms, which are often temperature-dependent.

3. Intelligent Monitoring and Alerts: An integrated system using both parameters can trigger real-time alerts during anomalies—such as spikes in water temperature—distinguishing between natural causes (e.g., heatwaves) and anthropogenic impacts (e.g., industrial discharge).

4. Optimized Environmental Regulation: In controlled ecosystems like aquaculture or urban water bodies, maintaining equilibrium between air and water temperature is crucial for species health and ecosystem balance. This dual data stream enables proactive environmental management

RESULTS

	A	B	C	D	E	F	G
1	Date	Time	Location	TDS (ppm)	Water Temp (°C)	Outside Temp (°C	pH
2	2025-04-01	8:00	Bandra West	210	26.3	29.5	7.1
3	2025-04-02	8:30	Andheri East	225	26.1	30	7.2
4	2025-04-03	9:00	Dadar	200	26.8	31.2	7
5	2025-04-04	8:15	Goregaon West	218	27	32.1	7.3
6	2025-04-05	8:45	Worli	230	26.5	30.9	7.1
7	2025-04-06	9:10	Borivali East	195	25.8	29.6	6.9
8	2025-04-07	8:00	Malad West	208	26.7	30.2	7
9	2025-04-08	8:30	Colaba	220	27.1	31	7.2
10	2025-04-09	9:00	Kurla	205	26	32.5	7.1
11	2025-04-10	8:15	Chembur	198	25.9	31.8	6.8
12	2025-04-11	8:40	Ghatkopar West	214	26.6	30.5	7.2
13	2025-04-12	9:05	Santacruz East	207	26.3	29.8	7
14	2025-04-13	8:10	Sion	193	25.5	30.7	6.9
15	2025-04-14	8:25	Matunga	202	26	31.1	7
16	2025-04-15	8:50	Khar Road	217	26.2	30.4	7.1
17	2025-04-16	9:20	Kandivali West	209	25.9	31.3	7
18	2025-04-17	8:00	Mazgaon	198	26.7	29.9	6.8
19	2025-04-18	8:35	Byculla	212	27.2	32.2	7.3
20	2025-04-19	9:00	Powai	220	26.5	31.7	7.1
21	2025-04-20	8:20	Mulund	204	25.7	30	6.9
22	2025-04-21	8:40	Bhandup	199	25.6	29.6	6.8
23	2025-04-22	9:10	Jogeshwari East	210	26.4	30.8	7
24	2025-04-23	8:05	Vikhroli	223	26.6	31.2	7.2
25	2025-04-24	8:30	Wadala	200	26	30.1	7
26	2025-04-25	9:00	Marine Lines	215	27	31.5	7.1
27	2025-04-26	8:10	Dahisar	197	25.5	29.7	6.9
28	2025-04-27	8:45	Charni Road	205	26.3	30.2	7
29	2025-04-28	9:15	Grant Road	221	26.8	30.9	7.2
30	2025-04-29	8:00	CST	208	26.1	31.3	7
31	2025-04-30	8:30	Sewri	203	26	30.5	6.9

	A	B	C	D	E	F	G	H
1	Date	Time	Location	TDS (ppm)	Water Temp (°C)	Outside Temp (°C)	pH	Water Quality Index
2	2025-04-01	8:00	Bandra West	210	26.3	29.5	7.1	Moderate
3	2025-04-02	8:30	Andheri East	225	26.1	30	7.2	Poor
4	2025-04-03	9:00	Dadar	200	26.8	31.2	7	Good
5	2025-04-04	8:15	Goregaon West	218	27	32.1	7.3	Moderate
6	2025-04-05	8:45	Worli	230	26.5	30.9	7.1	Poor
7	2025-04-06	9:10	Borivali East	195	25.8	29.6	6.9	Good
8	2025-04-07	8:00	Malad West	208	26.7	30.2	7	Moderate
9	2025-04-07	8:10	Majiwada	200	26	31	7.3	Good
10	2025-04-08	8:30	Colaba	220	27.1	31	7.2	Moderate
11	2025-04-08	8:15	Vashi	190	26.1	31.2	7.2	Good
12	2025-04-09	9:00	Kurla	205	26	32.5	7.1	Moderate
13	2025-04-09	7:50	Panvel	195	25.7	30	7.4	Good
14	2025-04-10	8:15	Chembur	198	25.9	31.8	6.8	Good
15	2025-04-10	8:42	Seawoods	210	26.4	30.8	7	Moderate
16	2025-04-11	8:40	Ghatkopar West	214	26.6	30.5	7.2	Moderate
17	2025-04-11	9:10	New Panvel	230	27.5	31.5	6.4	Poor
18	2025-04-12	9:05	Santacruz East	207	26.3	29.8	7	Moderate
19	2025-04-12	7:55	Kamothe	225	27	32.1	6.8	Poor
20	2025-04-13	8:10	Sion	193	25.5	30.7	6.9	Good
21	2025-04-13	8:20	Kalamboli	215	26.9	32.2	7.1	Moderate
22	2025-04-14	8:25	Matunga	202	26	31.1	7	Moderate
23	2025-04-14	9:00	Kopri	220	26.7	30.7	7	Moderate
24	2025-04-15	8:50	Khar Road	217	26.2	30.4	7.1	Moderate
25	2025-04-15	9:05	Sanpada	180	26.3	29.9	7.5	Good
26	2025-04-16	9:20	Kandivali West	209	25.9	31.3	7	Moderate
27	2025-04-16	8:50	Manpada	235	27.1	32.5	6.2	Poor
28	2025-04-17	8:00	Mazgaon	198	26.7	29.9	6.8	Good
29	2025-04-18	8:35	Byculla	212	27.2	32.2	7.3	Moderate
30	2025-04-18	7:45	Palaspe	185	26.2	30.1	7.6	Good
31	2025-04-19	9:00	Powai	220	26.5	31.7	7.1	Moderate

Figure 5.1 Data Collected

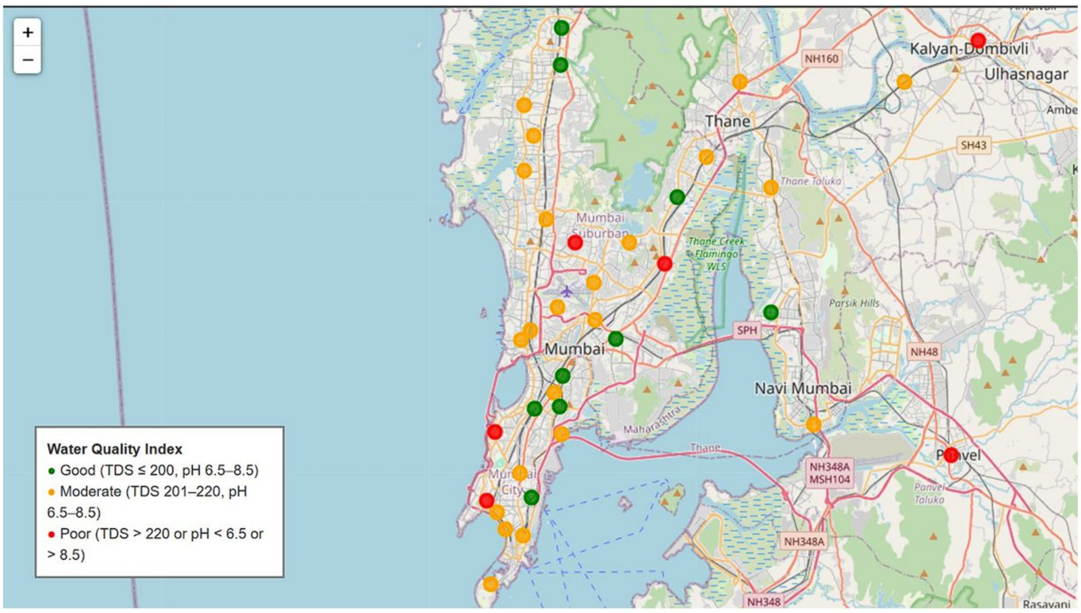

Figure 5.2 Analysis and Mapping

DISCUSSION

1. Good Quality Water (Green markers)

Found in areas like Dadar, Chembur, Borivali, Vashi, and Sion.

TDS ≤ 200 ppm and pH within 6.5–8.5 range — safest for drinking.

2. Moderate Water Quality (Orange markers)

Most common across the region.

Found in Ghatkopar, Marine Lines, Thane, Koparkhairane, and Turbhe.

TDS in 201–220 range with acceptable pH — may need filtration.

3. Poor Quality Water (Red markers)

Notable in Andheri East, Worli, Panvel, Ghansoli, and Kalyan.

Due to TDS > 220 or pH outside 6.5–8.5 — not suitable for direct consumption.

Geographic Observations:

Mumbai city has a higher mix of moderate and poor quality.

Navi Mumbai–Thane region mostly moderate, with a few poor hotspots.

Gaps filled with Koparkhairane, Rabale, Turbhe, Ghansoli show detailed transition in quality

FUTURE SCOPE

To build a powerful and intelligent water quality monitoring system using Raspberry Pi 4/5, capable of delivering fast, accurate, and optimized results. Our goal is to create an easy-to-use device, accessible to both literate and illiterate users, that leverages AI to classify water based on real-time sensor data.

We aim to:

1. Rework and optimize our prototype with advanced hardware and refined code.

2.Calibrate all sensors for maximum reliability across rural and industrial environments.

3. Build a scalable cloud infrastructure to manage and analyze nationwide sensor data in real-time. This system will form the backbone of a smart, accessible, and data-driven approach to clean water monitoring across India and beyond

4. Integrate high-accuracy sensors for parameters like Dissolved Oxygen, Ammonia, Nitrates, Heavy Metals, and Chlorine.

5. Deploy machine learning models (e.g., KNN, Random Forest) to estimate water quality even in unmapped or low-data areas.

CONCLUSION

The Water Quality Detection System using Raspberry Pi Pico successfully fulfills its objective of real-time monitoring of essential water quality parameters, namely pH level, Total Dissolved Solids (TDS), air temperature, and water temperature. By integrating low-cost, widely available sensors with the compact yet capable Raspberry Pi Pico microcontroller, the project delivers a reliable and affordable solution for basic water quality assessment. Its design emphasizes simplicity, practicality, and cost-effectiveness, making it highly suitable for educational settings, hobbyist use, and small-scale applications such as household water safety checks, hydroponic farming, aquaculture, and aquarium management.

This system demonstrates how embedded systems can be effectively applied to environmental sensing, providing valuable insights into water conditions that directly impact both human health and aquatic life. The clear and immediate feedback it offers empowers users to take timely action when water quality deviates from safe or optimal ranges. Through this project, key concepts of embedded electronics, sensor interfacing, and real-time data processing are not only showcased but also made approachable for beginners and enthusiasts interested in environmental monitoring technologies.

In addition to its current functionality, the project offers considerable potential for further development and scalability. Future improvements could include incorporating data logging features to track historical trends, adding wireless communication modules (such as Wi-Fi or Bluetooth) for remote monitoring, and integrating cloud services for advanced analytics and alerts. Enhancements like automated calibration routines, more sophisticated user interfaces, and expanded sensor arrays could also transform the system into a more comprehensive water quality monitoring solution suitable for broader applications.

In conclusion, this project serves as a strong foundation for anyone looking to explore the intersection of embedded systems and environmental sustainability. It effectively bridges the gap between theoretical learning and practical implementation, reinforcing the importance of technology-driven solutions in safeguarding natural resources. The success of this system underscores how simple yet innovative approaches can make meaningful contributions to both education and real-world problem-solving in the field of environmental monitoring.

APPENDIX

Program Code:

```
from machine import Pin, I2C, ADC
import time
import dht
from ds18x20 import DS18X20
import onewire

# Initialize I2C for LCD (if you're using I2C LCD)
i2c = I2C(0, scl=Pin(1), sda=Pin(0), freq=400000)

# Initialize DHT11 sensor for air temperature (GPIO 15)
dht_sensor = dht.DHT11(Pin(15))

# Initialize DS18B20 sensor for water temperature (GPIO 2)
ow = onewire.OneWire(Pin(2))
ds = DS18X20(ow)

# Initialize ADC for pH and TDS sensors
ph_sensor = ADC(Pin(26))   # pH sensor (GPIO 26)
tds_sensor = ADC(Pin(27))  # TDS sensor (GPIO 27)

# Function to read air temperature
def read_air_temperature():
    dht_sensor.measure()
    return dht_sensor.temperature()
```

```
# Function to read water temperature
def read_water_temperature():
    roms = ds.scan()
    ds.convert_temp()
    time.sleep(1)
    return ds.read_temp(roms[0])

# Function to read and calculate pH value
def read_ph_value():
    ph_raw = ph_sensor.read_u16()
    voltage = (ph_raw / 65535) * 3.3  # Convert to voltage (0-3.3V)
    # Replace with your calibrated formula
    ph_value = -5.70 * voltage + 20.0  # Example formula based on buffer calibration
    return ph_value

# Function to read and calculate TDS value (adjusted for Pico 3.3V ADC)
def read_tds_value():
    tds_raw = tds_sensor.read_u16()
    voltage = (tds_raw / 65535) * 3.3  # Convert to voltage
    # Clamp max voltage to 2.3V (sensor range)
    if voltage > 2.3:
        voltage = 2.3
    # Scale voltage from 3.3V to 2.3V range
    voltage = voltage * (2.3 / 3.3)
    # Apply Gravity TDS formula
    tds_value = (133.42 * voltage**3 - 255.86 * voltage**2 + 857.39 * voltage)
    return tds_value

# Main loop
while True:
```

```
air_temp = read_air_temperature()
water_temp = read_water_temperature()
ph_value = read_ph_value()
tds_value = read_tds_value()

print("Air Temperature: {}°C".format(air_temp))
print("Water Temperature: {:.2f}°C".format(water_temp))
print("pH Value: {:.2f}".format(ph_value))
print("TDS Value: {:.2f} ppm".format(tds_value))

# Optional LCD display (if using an LCD)
# lcd.clear()
# lcd.putstr("Air: {}C".format(air_temp))
# time.sleep(2)
# lcd.clear()
# lcd.putstr("Water: {:.1f}C".format(water_temp))
# time.sleep(2)
# lcd.clear()
# lcd.putstr("pH: {:.2f}".format(ph_value))
# time.sleep(2)
# lcd.clear()
# lcd.putstr("TDS: {:.0f}ppm".format(tds_value))

time.sleep(5)
```

IC DATASHEETS

1. Raspberry Pi Pico W (RP2040 Microcontroller):

https://datasheets.raspberrypi.com/picow/pico-w-datasheet.pdf

2. pH Sensor (HW-828):

https://www.electronicwings.com/sensors/ph-sensor-sk103-ph-sensor-module

3. TDS Sensor (V1.0):

https://wiki.dfrobot.com/TDS_Meter_Sensor_SKU__SEN0244

4. Temperature Sensor (DS18B20 Waterproof):

https://datasheets.maximintegrated.com/en/ds/DS18B20.pdf

5. JHD162A 16x2 I2C LCD Display:

https://www.jhdlcd.com/uploads/file/20180614/1528950470911796.pdf

I2C backpack: https://www.nxp.com/docs/en/data-sheet/PCF8574_PCF8574A.pdf

6. Breadboard & Jumper Wires:

(No datasheet needed, general-purpose hardware.)

7. Power Supply/USB Cable:

(No datasheet needed, general-purpose hardware.)

COST BREAKDOWN

Sr.No	Component	Quantity	Price(₹)	Total(₹)
1	Raspberry Pi Pico W	1	600	600
2	pH Sensor (HW-828)	1	800	800
3	TDS Sensor (V1.0)	1	900	900
4	DS18B20 Waterproof Temperature Sensor	1	300	300
5	External Humidity + Temp Sensor (DHT11)	1	100	100
6	JHD162A 16x2 I2C LCD Display (with I2C module)	1	300	300
7	Breadboard (good quality)	1	200	200
8	Jumper Wires (Male/Female mix, 65 pcs set)	1	120	120
9	USB Cable (Pico W power + Data)	1	120	120
10	Power Adapter (5V 2A, branded)	1	250	250
11	Miscellaneous (Resistors, Glue, Headers etc.)	1	100	100
TOTAL				≈₹4000

REFERENCES

pH Sensor HW-828 Datasheet :-
https://wiki.dfrobot.com/PH_meter_SKU__SEN0161_V2

TDS Sensor V1.0 (Analog TDS Sensor) : -
https://wiki.dfrobot.com/TDS_Sensor_SKU__SEN0244_

Raspberry Pi Pico W Documentation :-
https://www.raspberrypi.com/documentation/microcontrollers/

MicroPython :- https://micropython.org/

Raspberry Pi Pico MicroPython Guide :-
https://projects.raspberrypi.org/en/projects/getting-started-with-the-pico

ADS1115 ADC Module Datasheet:-
https://www.ti.com/product/ADS1115

Scikit-learn Documentation :-
https://scikit-learn.org/stable/documentation.html

TensorFlow Documentation :-
https://www.tensorflow.org/

MicroPython Libraries for Pico :-
https://docs.micropython.org/en/latest/library/index.html

Git & GitHub Documentation:-

https://git-scm.com/doc

https://docs.github.com/

Thonny IDE for MicroPython :-

https://thonny.org/